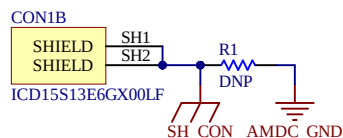


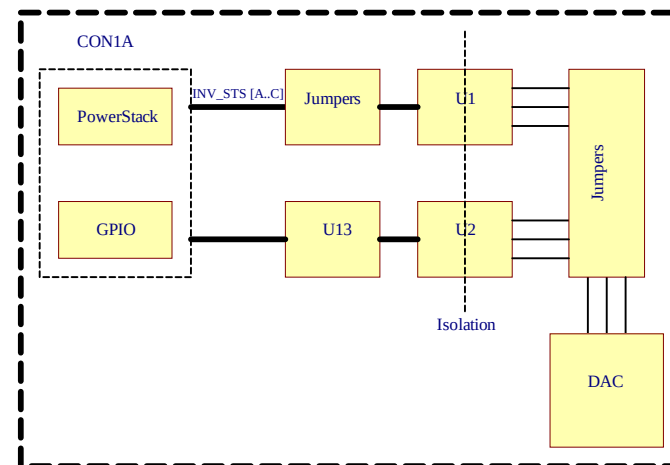
CON1 can be connected to both the power stack and GPIO connectors. Jumpers J1-J5 are used to configure the DAC between the powerstack and GPIO pinouts.

Legend:  
 ✓ Used  
 ✗ Unused

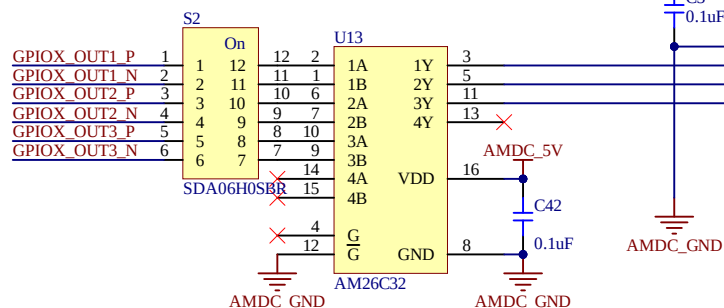
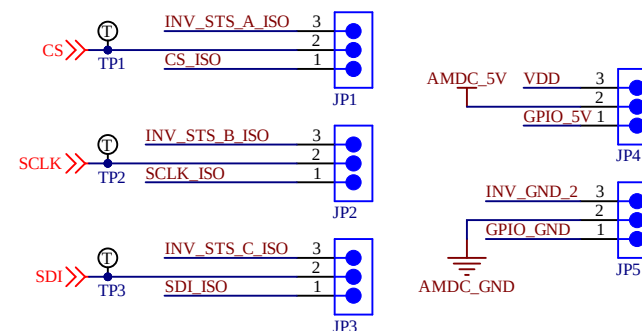
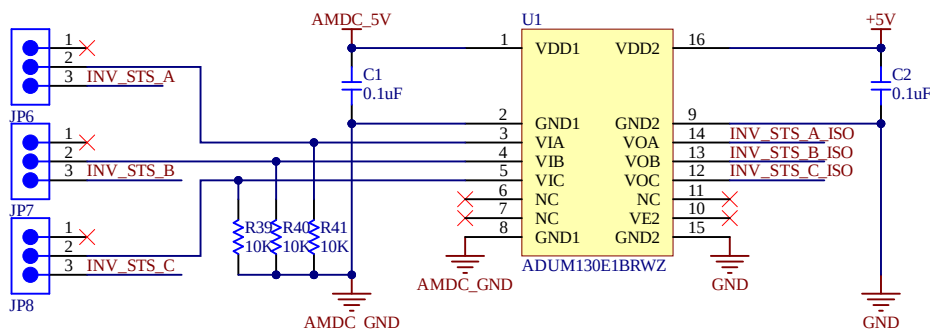
| Power Stack Pinout |   | GPIO Pinout  |   |
|--------------------|---|--------------|---|
| CON1A              |   |              |   |
| 1                  | ✗ | INV_PWM_1    | ✗ |
| 2                  | ✗ | INV_PWM_2    | ✗ |
| 3                  | ✗ | INV_PWM_3    | ✗ |
| 4                  | ✗ | INV_PWM_4    | ✗ |
| 5                  | ✗ | INV_PWM_5    | ✗ |
| 6                  | ✗ | INV_PWM_6    | ✗ |
| 7                  | ✗ | INV_GND_1    | ✗ |
| 8                  | ✗ | INV_STS_A    | ✓ |
| 9                  | ✗ | INV_STS_B    | ✓ |
| 10                 | ✗ | INV_STS_C    | ✓ |
| 11                 | ✗ | INV_STS_D    | ✓ |
| 12                 | ✗ | VPS          | ✗ |
| 13                 | ✗ | VDD          | ✓ |
| 14                 | ✗ | PSGND        | ✓ |
| 15                 | ✗ | INV_GND_2    | ✓ |
| ICD15S13E6GX00LF   |   |              |   |
| 1                  | ✗ | GPIOX_IN1_P  | ✗ |
| 2                  | ✗ | GPIOX_IN1_N  | ✗ |
| 3                  | ✗ | GPIOX_IN2_P  | ✗ |
| 4                  | ✗ | GPIOX_IN2_N  | ✗ |
| 5                  | ✗ | GPIOX_IN3_P  | ✗ |
| 6                  | ✗ | GPIOX_IN3_N  | ✗ |
| 7                  | ✗ | GPIOX_OUT3_P | ✓ |
| 8                  | ✗ | GPIOX_OUT3_N | ✓ |
| 9                  | ✗ | GPIOX_OUT1_P | ✓ |
| 10                 | ✗ | GPIOX_OUT1_N | ✓ |
| 11                 | ✗ | GPIOX_OUT2_P | ✓ |
| 12                 | ✗ | GPIOX_OUT2_N | ✓ |
|                    |   | GPIO_GND     |   |

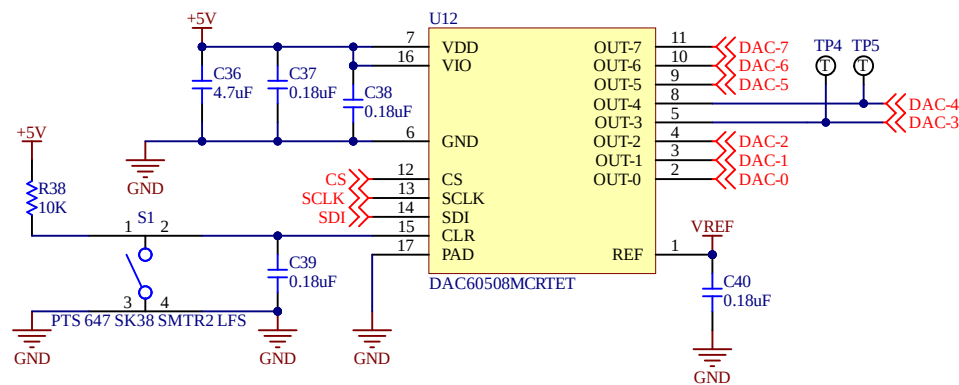
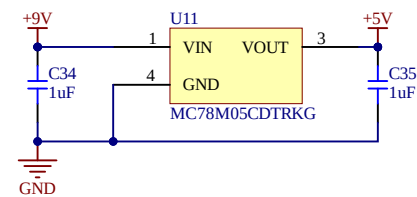
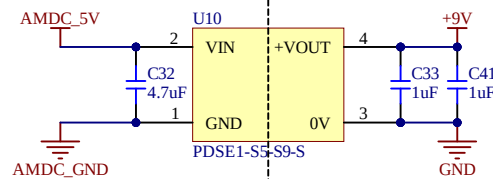
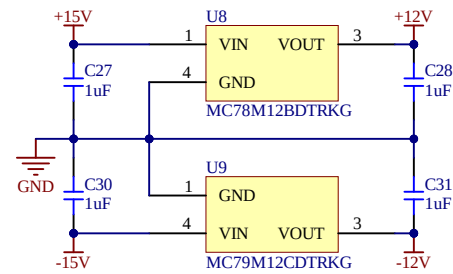
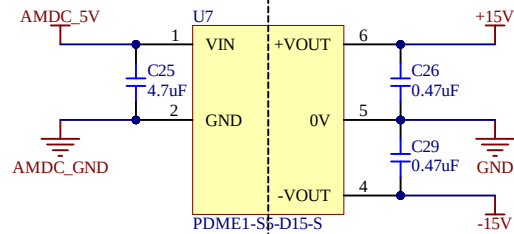


Connector Signal Diagram



Jumpers J6-J8 are used to disable U1 when the power stack is not in use



Title: **DAC**

File: DAC.SchDoc

Revision: D

Time: 4:59:24 PM

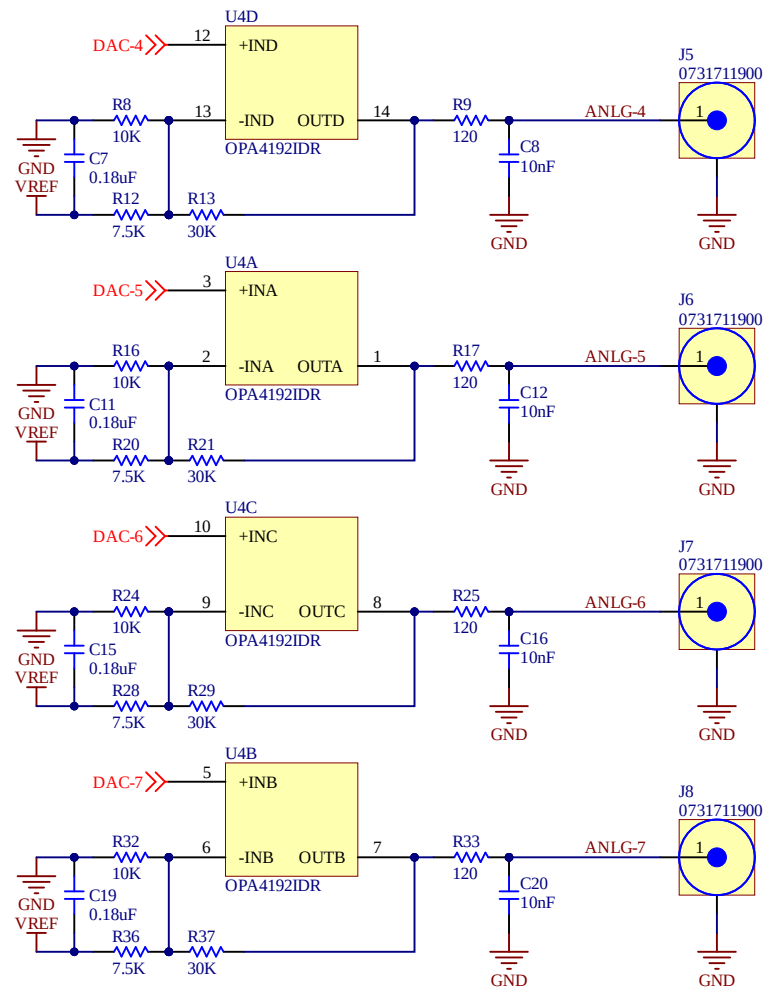
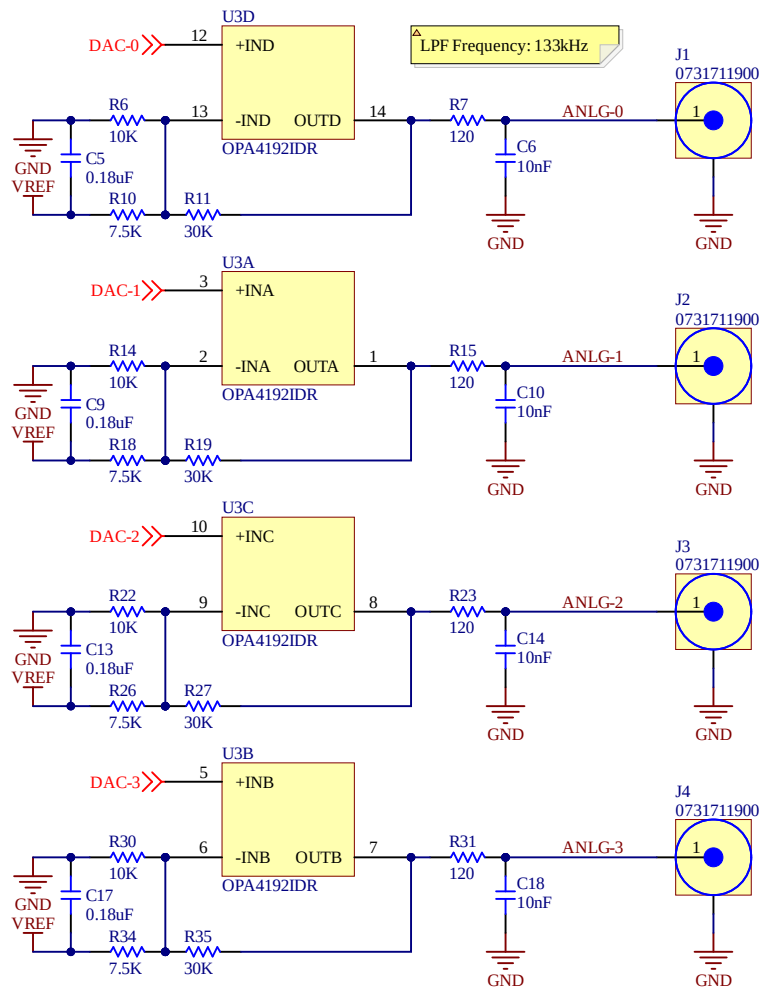
Sheet: 2 of 4

Date: 7/6/2026

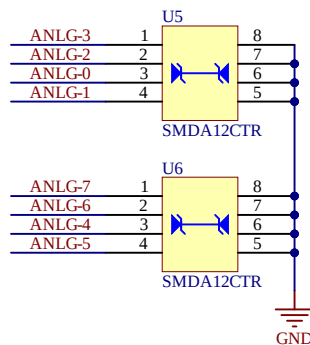
Severson Group  
WEMPEC  
UW-Madison

Engineer: Brandt Buchda

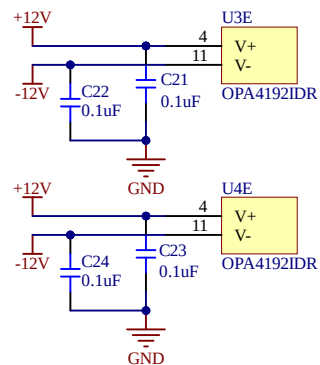




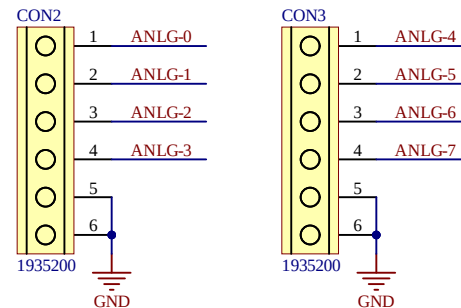
## ESD Protection



## OP Amp Power Supply



## Screw Terminal Connectors

Title: **Output Connectors**

File: OutputConnectors.SchDoc

Revision: D

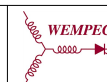
Time: 4:59:24 PM

Sheet: 3 of 4

Date: 7/6/2026

Severson Group  
WEMPEC  
UW-Madison

Engineer: Brandt Buchda &amp; Patrick Nowakowski



4-40 Screws:

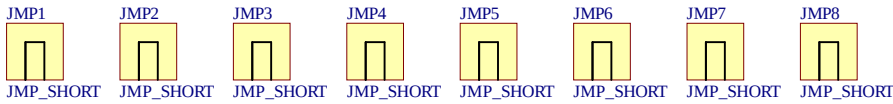


4-40 Standoffs:



Pinout Jumpers:

Silkscreen marked with default locations



Mounting Holes:

